



Artifact - The Dota card game

Valve released a new 35-second teaser video of a new card game, set to launch sometime in 2018.
<http://digit.in/ArtifactD2>



Bad news, Middle Earth fans

It was recently announced that Middle Earth: Shadow of War will have microtransactions.
<http://digit.in/LOTRcash>

light' (FTL) speed explained? The dark matter moves the universe around the ship, instead of the ship itself! Another notable example of a dark matter drive is from Space Pirate Captain Harlock, a manga-inspired story where dark matter engines grant the ship Arcadia the ability to travel at a massively FTL speed (and immensely powerful destructive abilities).



Space Pirate Captain Harlock is a must watch for sci-fi fans

Interestingly, this idea isn't as unreal as one would expect. An NYU physicist Jia Liu had proposed a spaceship design that would scoop up dark matter particles at the front. According to the theory that dark matter is made up of neutralinos, the particles would annihilate each other and provide sufficient energy, if directed correctly, to reach lightspeed in a matter of days. Of course, this involves a ton of assumptions, as does any dark matter theory.

SOLAR/STAR POWERED PROPULSION

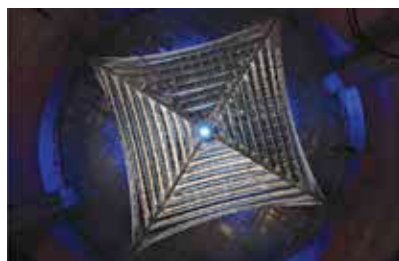
Solar power in space is used in a number of orbital spacecrafts and satellites. But to use solar power as a source of propulsion is far fetched, even for science fiction. The area of solar cells required, combined with the size of the batteries needed to store the solar power are enough to render it unfeasible as a sole source of power in our own solar system. A higher density of stars, however, does make this a possibility. A more popular form of star-based power in science fiction is harnessed by using solar sails.

Since James Clarke Maxwell discovered that light exerts a certain amount of pressure on objects it falls

upon, the idea of solar sails propelling spacecrafts has captured the imagination of many creative minds. Initially appearing in Jules Verne's 1865 novel From the Earth to the Moon, the idea got a strong boost in popularity with Arthur C. Clarke's Sunjammer almost a hundred years later. In fact, NASA's first solar sail mission to deep space has been dubbed "Sunjammer" in Clarke's honor.

Other notable examples include the Bajoran lightship from Star Trek: Deep Space Nine and Count Dooku's solar sailer from Star Wars Episode II: Attack of the Clones. And just like the others, even this idea has found a real counterpart ever since the beginning of the space race.

Lightsail 2, the successor to the Planetary Society's Lightsail-1 (formerly Lightsail-A), is set to be launched aboard the first operational SpaceX Falcon Heavy in 2018 to test the viability of solar sails as a propulsion mechanism for CubeSats. Additionally, in 2010, IKAROS (Interplanetary Kite-craft Accelerated by Radiation Of the Sun) was launched as the first spacecraft to use solar sails as a primary propulsion system. And there are even more projects planned in the coming years.



The Sunjammer solar sail prototype

AND THEN THERE ARE THESE

While most spacecrafts in sci-fi try to base their propulsion mechanism in current scientific advancements, there are some that have redefined what propulsion mechanisms can do and used entirely out of the galaxy thinking to travel across space like never before.

If you're a true geek, it's highly likely that you are familiar with Hitchhiker's Guide to the Galaxy. The iconic

Heart of Gold spacecraft in the series uses an Infinite Improbability Drive to travel at, well, infinite speed (what else could it be, right?). According to the book, the drive works on a combination of a Finite Improbability Generator and a cup of tea with a little bit of Brownian motion thrown in. This allows the drive, and the ship it powers, to travel through every conceivable point in every conceivable universe simultaneously every time it is used, making it the fastest ship possible in science fiction.



The iconic blue police box that's the Doctor's ship

Any geek worth his money knows about Doctor Who and Tardis. The space-time transcending police box that the Time Lord travels in is actually a type 40 time and space machine from Gallifrey, his home planet. The ship, if we may call it so, derives its powers from multiple sources. One of the primary ones is the Eye of Harmony, which is the nucleus of a black hole created by early Time Lords - essentially, a singularity. This allows the ship to travel through the time vortex without actually travelling through space, although the TARDIS is shown to fly on several occasions as well.

ONBOARD

At this point, it wouldn't be entirely wrong to compare the volume of space-based science fiction out there with space itself. Hence, it might have been fairly impossible on our part to cover every possible type of popular spacecraft-propelling drives and engines in sci-fi. Although, if you think we missed out a particularly significant one, do shoot us an email at editor@digit.in! 📧